

Which Metric, and Why: Premium-Listing Classifier

Model: Logistic regression, features = age + miles, held out 20% of listings, ~8–10% premium base rate. ROC-AUC 0.928.

Confusion matrix — default (0.50) threshold

	Pred. regular	Pred. premium
Actual regular	1,711 (TN)	9 (FP)
Actual premium	117 (FN)	27 (TP)

Accuracy 0.932, Precision 0.750, Recall 0.188, F1 0.300

Confusion matrix — chosen (0.25) threshold

	Pred. regular	Pred. premium
Actual regular	1,618 (TN)	102 (FP)
Actual premium	62 (FN)	82 (TP)

Accuracy 0.912, Precision 0.446, Recall 0.569, F1 0.500

Metric chosen: Recall

Accuracy is discarded immediately on this base rate, a model that never predicts "premium" would score over 90% while being useless. The real decision is between recall, precision, and F1, and it depends on one business assumption stated here explicitly: flagging a listing "premium" triggers something cheap and reversible better photos, a featured placement, a different price suggestion not an expensive or binding action. Under that assumption, a false positive costs almost nothing; a false negative is a quiet, permanent loss, since a missed premium car sits under-priced or under-marketed for its whole time on the platform.. If that assumption is wrong if the flag instead triggers something costly, like a manual appraisal or a pricing commitment the conclusion flips toward precision instead. That is a business call, not a modeling call, and it should be stated rather than hidden inside a metric like F1 that quietly treats both error types as equally costly.

Why the default threshold is the wrong baseline

At the default 0.50 cutoff, recall is only 0.188 — the model misses over 80% of premium cars. That looks like a weak model, but ROC-AUC of 0.928 shows it actually ranks premium listings well above regular ones. The problem is not the model; 0.50 was never chosen for this objective, it is simply an untuned default. The fix is not a better model — it is a deliberately chosen threshold. At 0.25, recall rises to 0.569 (more than triple), while precision falls to 0.446, meaning roughly 55% of "premium" flags are false alarms.

